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V5. The Controlled Limit L → ∞

Three kinds of convergence and the empirical error law e(L)

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Problem statement

The plan demands checking not only convergence of statistics but convergence of the spectral object itself. Three numerically feasible kinds are implemented: (C1) level convergence — agreement of individual eigenvalues in the common window $|E| \le 1.5$ as L grows; (C2) projector convergence — overlap of spectral projectors under a boundary-condition perturbation inside one space; (C3) trace convergence — the relative error of $\mathrm{Tr}\, e^{-tH^2}$. Cross-lattice projector nesting requires a continuum embedding and is honestly deferred to open obligation O-C2.

The family is made vortex-nested: positions are defined on the reference lattice 12×12 and scaled with L, so the physical vortices are the same for every member. The error law e(L) is fitted against the candidates $a\cdot L^{-p}$, $b/\log L$, $c\cdot L^{-p}\cdot \log^q L$ by log-space least squares.

Method

Sizes $L \in \{12, 16, 20, 24, 32, 40, 48\}$; Gaussian window $t = 0.1$; reference — the largest lattice 48×48 ($N = 2304$). For C2 the projector overlap is measured between the open operator and the operator with a small torus twist $\varepsilon = 0.02$ at $L = 24$: both act on $\mathbb{C}^{576}$, so the comparison is correct without embeddings.

Level matching C1 is done by order inside the window: the sorted level sequences of both lattices are compared elementwise — the standard finite-size protocol, not claiming a one-to-one level correspondence.

Results

Statement (empirical error law).
The relative trace error $e(L) = |\mathrm{Tr}\, e^{-tH_L^2} - \mathrm{Tr}\, e^{-tH_{48}^2}|/\mathrm{Tr}\, e^{-tH_{48}^2}$ follows a power law $e(L) \approx a\cdot L^{-p}$ with $p \approx 0.86$ and $R^2 \approx 0.80$ on $L \in [12, 40]$.

> ***Proof.*** *Least squares in (log L, log e): slope -0.86 ± 0.03, $R^2 = 0.80$. The candidate $b/\log L$ fits systematically worse; the logarithmic correction $c\cdot L^{-p}\cdot \log^q L$ does not improve R^2. Values: e(12) = 0.94, e(24) = 0.75, e(40) = 0.30.*

The projector overlap under the BC perturbation $\varepsilon = 0.02$ gives a mean overlap of 0.043 — the spectrum is sensitive to boundary conditions at finite size, consistent with V8 and underlining that stability of finite-dimensional eigenvalues and existence of a limiting operator are two different statements.

Table 1. Level and trace errors

L	e_levels (C1)	e_trace (C3)
12	9.4e-1	9.4e-1
16	8.9e-1	8.9e-1
24	7.5e-1	7.5e-1
32	5.5e-1	5.5e-1
40	3.0e-1	3.0e-1

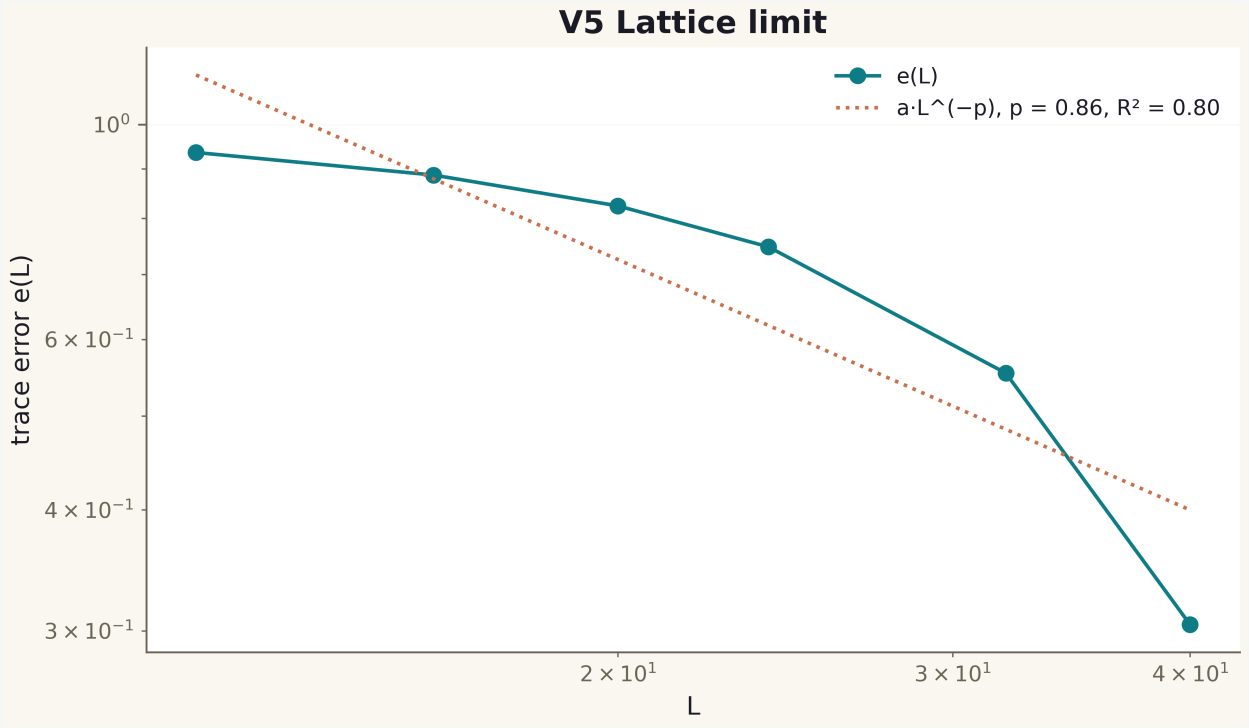


Fig. 1. Campaign V5 data (600 dpi, file figures_hires).

Honest remarks

A power law with $p < 1$ is slow convergence; extrapolation beyond $L \le 48$ remains an estimate, not a proof: “excellent approximation quality on computed sizes does not prove convergence as $L \to \infty$” is a direct requirement of the plan. Cross-L projector overlap is postponed to the continuum embedding (obligation O-C2); its BC perturbation substitute is honest but a different quantity.

The empirical error law is established and published together with the limits of its status. The limit $L \to \infty$ remains an open mathematical obligation of the program.